Fundamentals of Data Collection
SURV 621/SURVMETH 621
Fall 2024
(Last updated: 09/30/2024)

Monday 1:00-4:00pm

University of Maryland – 1208 LeFrak Hall
University of Michigan – 1070 ISR

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Fundamentals of Data Collection is a two-semester course sequence that provides a broad overview of the processes that generate data – data that are *designed*, primarily survey data, and those that are *found*, e.g., administrative records, smartphone sensor data, and social media posts. The course exposes students to key theoretical ideas about data collection through lectures that are supplemented by readings drawn from the relevant scientific literature. The course provides practical experience through exercises conducted mostly outside of class. Students will gain a thorough understanding of a wide range of data as well as their relative strengths and weaknesses. The first semester focuses more on data from surveys than alternative sources; the second semester reverses this focus.

The first semester begins with a module that introduces the Total Survey Error framework and considers its extension to found data. The second module explores sources of representation error, such as coverage error and nonresponse error, in both designed and found data. The following module examines sources of measurement error and considers how to write survey questionnaires that minimize such errors. The fourth module concerns modes of collecting survey data, both classic (mail, face-to-face, telephone and web) and emerging (e.g., text/SMS and video) modes, as well as mixed mode surveys. The fifth module examines self-administered survey modes, particularly online (web) surveys, in greater detail and the sixth module focuses on interviews and interviewers.

Course Requirements:
Reading assignments for each class are available on the course website (https://umich.instructure.com/courses/700521). Please read the assigned material for each class session beforehand.

Grades for the course will be based on:

- Participation in class demonstrating understanding of the readings and in-class material (10% of grade). You are expected to post a comment or question about one of the readings to the Canvas Discussion Forum before each class. Students are encouraged to respond to each other's posts.

- Practical exercises (60% of grade)
- Midterm exam (15% of grade)
- Final exam (15% of grade)

Note: For practical exercises carried out in groups, the members of each group will rate each of the other members' contributions to the assignment; average ratings will be used to adjust the score each student receives for their work on the assignment

Attending Class

You are expected to attend class in person unless extenuating circumstances prevent this. In such cases and with the instructor's approval you can attend that class via Zoom. You must obtain the instructor's approval to attend class remotely for any class you are unable to attend in person. One exception: Some JPSM students have been approved to attend classes remotely and so will attend all classes via Zoom.

You may wish to join the recurring Zoom "meeting" for a class from within the classroom. If so, be sure to mute your computer or other device. The link for the recurring Zoom meeting is: https://umich.zoom.us/j/94206859614 passcode: 700521. If you are approved to join via Zoom remotely, you are free to use a virtual background. Please be sure to mute your microphone when you are not speaking. You will need a good Internet connection and may need a headset including microphone for the experience to be positive and effective. If you need assistance connecting to or testing Zoom, please contact Elisabeth Schneider (dodo@umich.edu) or Patsy Gregory (pagregor@umich.edu).

Accommodations for Students with Disabilities

University of Michigan

If you believe you need an accommodation for a disability, please contact the Services for Students with Disabilities (SSD) office to help us determine appropriate academic accommodations. SSD (734-763-3000; http://ssd.umich.edu) typically recommends accommodations through a Verified Individualized Services and Accommodations (VISA) form. Any information you provide is private and will remain confidential.

University of Maryland

If you believe you need an accommodation for a disability, please contact the Accessibility and Disability Services (ADS) office to register for services: 301-314-7682. More information is available at https://counseling.umd.edu/ads.

Academic conduct

Clear definitions of the forms of academic misconduct, including cheating and plagiarism, as well as information about disciplinary sanctions for academic misconduct at the University of Michigan may be found at the Rackham web site (https://rackham.umich.edu/academic-policies/section8/). The analogous information for the University of Maryland can be found at the web site for the Office of the President (https://www.president.umd.edu/sites/president.umd.edu/files/documents/policies/III-100A.pdf).

Knowledge of these rules is the responsibility of the student and ignorance of them does not excuse misconduct. The student is expected to be familiar with these guidelines before submitting any written work or taking any exams in this course. Lack of familiarity with these rules in no way constitutes an excuse for acts of misconduct. Charges of plagiarism and other forms of academic misconduct will be dealt with very seriously and may result in oral or written reprimands, a lower or failing grade on the assignment, a lower or failing grade for the course, suspension, and/or, in some cases, expulsion from the university.

Schedule and Reading Assignments

FALL SEMESTER

Module 1: Total Survey Error in Designed and Found Data

August 26: Introduction to course (Antoun); introduction to Total Survey Error framework (Antoun)
No discussion forum this week
Exercise 1 distributed

Brackstone, G. 1999. "Managing data quality in a statistical agency," *Survey Methodology* 25(2), 139-149.

Groves, R. et al. 2009. "Chapter 2: Inference and errors in surveys," *Survey Methodology 2nd Edition*, Hoboken, NJ: Wiley, 39-64.

Couper, M. P. 2013. "Is the sky falling? New technology, changing media, and the future of surveys, *Survey Research Methods*, 7(3), 145-156.

Baker, R. 2017. "Big data: A survey research perspective," in P. Biemer et al. (eds). *Total Survey Error in Practice*, Hoboken, NJ: Wiley, 47-70.

Optional readings:

Citro, C. 2014. "From multiple modes for surveys to multiple data sources for estimates," *Survey Methodology* 40(2), 137-161.

Daas, P.J.H. et al. 2015. "Big data as a source for official statistics," *Journal of Official Statistics* 31(2), 249-262.

Salanik, M.J. 2018. "Chapter 2: Observing Behavior," *Bit by Bit: Social Research in the Digital*

Age, Princeton, NJ: Princeton University Press, 13-62. (https://www.bitbybitbook.com/en/1st-ed/observing-behavior/)

September 2: No class; Labor Day

Module 2: Errors of Representation in Designed and Found Data

September 9: Coverage and nonresponse error in probability surveys (Antoun)

Groves, R. et al. 2009. "Chapter 3: Target populations, sampling frames and coverage error," *Survey Methodology 2nd Edition*, Hoboken, NJ: Wiley, 69-95.

Groves, R. et al. 2009. "Chapter 6: Nonresponse in sample surveys," *Survey Methodology 2nd Edition*, Hoboken, NJ: Wiley, 183-211.

Groves, R. and E. Peytcheva. 2008. "The impact of nonresponse rates on nonresponse bias: A meta-analysis," *Public Opinion Quarterly*, 72(2), 167-189.

Optional readings:

Link, M., M. Battaglia, M. Frankel, L. Osborn and A. Mokdad. 2008. "A comparison of address-based sampling (ABS) versus random-digit dialing (RDD) for general population surveys," Public Opinion Quarterly, 72(1), 6–27.

Keusch, F. (2015). Why do people participate in Web surveys? Applying survey participation theory to Internet survey data collection. *Management review quarterly*, *65*(3), 183-216.

Martin, E. 1999. "Who knows who lives here? Within-household disagreements as a source of coverage error," *Public Opinion Quarterly*, 63(2), 220-236.

American Association for Public Opinion Research (AAPOR). 2016. *Standard Definitions: Final Dispositions of Case Codes and Outcome Rates for Surveys*.

Federal Committee on Statistical Methodology. 2001. "Chapter 4: Nonresponse error," in *Measuring and Reporting Sources of Error in Surveys*, Statistical Policy Working Paper No. 31, 4-1 to 4-23.

Abraham, K., A. Maitland and S. Bianchi. 2006. "Nonresponse in the American Time Use Survey: Who is missing from the data and how much does it matter," *Public Opinion Quarterly,* 70(5), 676–703.

September 16: Representation error in nonprobability surveys and non-survey data (Antoun)

Cornesse, C., Blom, A. G., Dutwin, D., Krosnick, J. A., De Leeuw, E. D., Legleye, S., ... & Wenz, A. (2020). A review of conceptual approaches and empirical evidence on probability and nonprobability sample survey research. *Journal of Survey Statistics and Methodology*, *8*(1), 4-36.

MacInnis, B., Krosnick, J. A., Ho, A. S., & Cho, M. J. (2018). The accuracy of measurements with probability and nonprobability survey samples: Replication and extension. *Public Opinion Quarterly*, *82*(4), 707-744.

AAPOR. 2013. "Chapter 6. Estimation and weight adjustment methods." pp 61-76. Rep*ort of the AAPOR task force on nonprobability sampling.*

Optional readings:

Yeager, D. S., Krosnick, J. A., Chang, L., Javitz, H. S., Levendusky, M. S., Simpser, A., & Wang, R. 2011. Comparing the accuracy of RDD telephone surveys and internet surveys conducted with probability and non-probability samples. Public opinion quarterly, 75(4), 709-747.

Schober, M. F., J. Pasek, L. Guggenheim, C. Lampe, and F.G. Conrad. 2016. "Social media analyses for social measurement," Public Opinion Quarterly, 80(1), 180-211.

Keusch, F., & Kreuter, F. 2021. Digital trace data: Modes of data collection, applications, and errors at a glance. In Handbook of Computational Social Science, Vol 1. Taylor & Francis.

Gelman, et al. 2014. "Harnessing naturally occurring data to measure the response of spending to income," Science, 345(6193), 212-215.

Valliant, R. and J. Dever. 2011. "Estimating propensity adjustments for volunteer web surveys," *Sociological Methods and Research*, 40(1), 105–137.

<u>Module 3: Measurement Error and Its Reduction</u>

September 23: Psychological origins of survey measurement error (Suzer-Gurtekin)

Exercise 1 due by 1PM

Tourangeau, R. and N. M. Bradburn. 2010. "Chapter 10: The psychology of survey response," in P.V. Marsden and J. D. Wright (eds.), *Handbook of Survey Research, 2nd Edition*, Emerald Group Publishing, 315-346.

Krosnick, J. A. 1999. "Survey research," *Annual review of psychology 50*(1), 537-567.

Sudman, S., N. M. Bradburn, and N. Schwarz. 1996. "Chapter 5: The direction of context effects: What determines assimilation or contrast in attitude measurement?" in *Thinking about answers: The application of cognitive processes to survey methodology*, Jossey-Bass,100-129.

Optional readings:

Krosnick, J. A. 1991. "Response strategies for coping with the cognitive demands of attitude measures in surveys." *Applied cognitive psychology*, *5*(3), 213-236.

Schwarz, N., F. Strack, and H.P. Mai. 1991. "Assimilation and contrast effects in part-whole question sequences: A conversational logic analysis." *Public opinion quarterly, 55*(1), 3-23.

Sudman, S., N. M. Bradburn, and N. Schwarz. 1996. "Chapter 4: Psychological sources of context effects in survey measurement" in *Thinking about answers: The application of cognitive processes to survey methodology*, Jossey-Bass,80-99.

September 30: Cognitive interviewing (Paul Beatty [guest lecturer], Suzer-Gurtekin)
Exercise 2 available at 2:30 on October 7

Beatty, P. C., and G.B. Willis. 2007. "Research synthesis: The practice of cognitive interviewing," *Public Opinion Quarterly*, *71*(2), 287-311.

Willis, G. 2004. "Cognitive Interviewing: A 'How To' Guide," unpublished report.

October 7: Reducing measurement error (Suzer-Gurtekin)
Midterm Exam available after class; due by 1:00 PM on October 21

Krosnick, J. A., and S. Presser. 2010. "Chapter 9: Question and questionnaire design," in P.V. Marsden and J. D. Wright (eds.) *Handbook of Survey Research, 2nd Edition*, Emerald Group Publishing, 263-314.

Fowler, F. J. 1995. "Chapter 2: Designing Questions to Gather Factual Data" in *Improving survey questions: Design and evaluation*, Thousand Oaks: Sage Publication, 8-45.

Fowler, F. J. 1995. "Chapter 3: Questions to Measure Subjective States" in *Improving survey questions: Design and evaluation*, Thousand Oaks: Sage Publication, 46-77.

Tourangeau, R. and T. Yan. 2007. "Sensitive questions in surveys," *Psychological Bulletin*, 133(5), 859.

Optional readings:

Liu, M., S. Lee, and F.G. Conrad. 2015. "Comparing extreme response styles between agree-disagree and item-specific scales." *Public Opinion Quarterly, 79*(4), 952-975.

October 14: No Class; Fall Study Break (U-M)

October 21: Probability-based web panels (Ipek Bilgen [guest lecturer]); Measurement beyond the (Household) Survey Interview (Suzer-Gurtekin)
Midterm Exam due by 1:00 PM on October 21

Marwick, A. E., and d. boyd. 2011. "I tweet honestly, I tweet passionately: Twitter users, context collapse, and the imagined audience." *New media & society 13* (1): 114-133.

Iacus, S. M., & Porro, G. (2022). Don't ask, just listen… Using social networks to measure subjective well-being. *Significance*, *19*(3), 10-15.

Optional readings:

Willimack, D. K. and E. Nichols. 2010. "A hybrid response process model for business surveys," *Journal of Official Statistics*, 26(1), 3–24.

Biemer, P. and R. S. Fecso. 1995. "Evaluating and controlling measurement error in business surveys," in B. Cox et al (eds), *Business Survey Methods*, New York: John Wiley and Sons, 257-281.

Smith, B. K., and Abel Gustafson. "Using Wikipedia to predict election outcomes: Online behavior as a predictor of voting." *Public Opinion Quarterly 81*, no. 3 (2017): 714-735.

Module 4: Introduction to Survey Mode

October 28: Classic survey modes; mode effects (Suzer-Gurtekin)
Exercise 2 due by 1PM
Exercise 2 peer assessment due
Exercise 3 available at 4PM

Biemer, P.P. and L.E. Lyberg. 2003. "Chapter 6: Data collection modes and associated errors," in *Introduction to Survey Quality*, New York: Wiley, 188-214.

Couper, M. P. 2011. "The future of modes of data collection," *Public Opinion Quarterly 75*(5), 889-908.

Kreuter, F., S. Presser and R. Tourangeau. 2008. "Social desirability bias in CATI, IVR, and web surveys: The effects of mode and question sensitivity," *Public Opinion Quarterly, 72,* 847-865.

Optional readings:

Module 4.1: Self-administration

November 4: Web surveys; designing questionnaires for online administration (Antoun)

Couper, M. 2000. "Web surveys: A review of issues and approaches," *Public Opinion Quarterly,* 64(4), 464-494.

AAPOR. 2010. "Chapter 3: An overview of online panels" and "Chapter 4: Errors of nonobservation in online panel surveys", *AAPOR report on online panels,* 9-28.

Tourangeau, R. Conrad, F., & Couper M. (2013). Chapter 4: Introduction to measurement and design in web surveys. The Science of Web Surveys. New York: Oxford University Press, 57-76.

Optional readings:

Couper, M. P. (2008). Chapter 2: The basic building blocks. Designing effective web surveys. Cambridge University Press, 41-83.

Snow, Jonathan. Undated. “The complete Research Suite: A step-by-step guide to using Qualtrics,” unpublished mimeograph.

November 11: Mobile web and text/SMS surveys (Antoun)

Conrad, F.G., M.F. Schober, A.L. Hupp, C. Antoun, and H.Y. Yan. 2017. “Text interviews on mobile devices,” in P. Biemer et al (eds), *Total Survey Error in Practice*, Hoboken, NJ: John Wiley & Sons, 299-318.

Callegaro, M., Lozar Manfreda, K., & Vehovar, V. (2015). Chapter 5.1: Smartphones, tablets and other devices. Web Survey Methodology. London: Sage, 192-205.

Optional readings:

Schober, M. F., Conrad, F. G., Antoun, C., Ehlen, P., Fail, S., Hupp, A. L., ... & Zhang, C. (2015). Precision and disclosure in text and voice interviews on smartphones. *PloS one*, *10*(6), e0128337.

Antoun, C., M. P. Couper, and F. G. Conrad. 2017. "Effects of mobile versus pc web on survey response quality: a crossover experiment in a probability web panel*." Public Opinion Quarterly* 81(S1), 280-306.

Couper, Mick P., Christopher Antoun, and Aigul Mavletova. 2017. "Mobile Web Surveys," in P. Biemer et al (eds), *Total Survey Error in Practice*, Hoboken, NJ: John Wiley & Sons, 133-154.

Lugtig, P. and V. Toepoel. 2016. “The use of PCs, smartphones, and tablets in a probability-based panel survey: Effects on survey measurement error,” *Social Science Computer Review*, 34, 78-94.

Mavletova, A. and M.P. Couper. 2014. “Mobile Web survey design: Scrolling versus paging, SMS versus e-mail invitations,” *Journal of Survey Statistics and Methodology*, 2, 498-518.

November 18: Self-administration in interviews and other mixed mode designs (Suzer-Gurtekin)

Exercise 3 due by 1PM
Exercise 3 peer assessment due
Exercise 4 available

Tourangeau, R. and T. W. Smith. 1996. "Asking sensitive questions: The impact of data collection mode, question format, and question context." *Public opinion quarterly 60*, (2) 275-304.

Steiger, D. M, and B. Conroy. 2008. "IVR: interactive voice response," in de E.D. Leeuw, J.J. Hox, D.A. Dillman (eds.*), International handbook of survey methodology*, Routledge. 285-98.

De Leeuw, E.D. 2018. "Mixed mode: Past, present and future." *Survey Research Methods, 12,* 75-89.

Optional readings:

Aquilino, W. S., D. L. Wright, and A. J. Supple. 2000. "Response effects due to bystander presence in CASI and paper-and-pencil surveys of drug use and alcohol use." *Substance Use & Misuse 35*(6-8), 845-867.

Conrad, F.G., M.F. Schober, C. Antoun, C., H. Y. Yan, A. L. Hupp, M. Johnston, P. Ehlen, L. Vickers, C. Zhang. 2017. "Respondent mode choice in a smartphone survey," *Public Opinion Quarterly,* 81, 307-337.

Couper, M.P., E. Singer, and R. Tourangeau, 2004. "Does voice matter? An interactive voice response (IVR) experiment." *Journal of Official Statistics*, 20(3), 551-570.

DeLeeuw, E. D. 2005. "To mix or not to mix data collection modes in surveys," *Journal of Official Statistics*, 21, 233-255.

Gnambs, T. and K. Kaspar, 2015. "Disclosure of sensitive behaviors across self-administered survey modes: a meta-analysis." *Behavior research methods*, *47*(4), 1237-1259.

Levine, B., Krotki, K. & Lavrakas, P. (2019). Redirected Inbound Call Sampling (RICS): Telephone surveying via a new survey sampling paradigm. *Public Opinion Quarterly.*

Lind, L.H., M.F. Schober, F.G. Conrad, and H. Reichert, 2013. "Why do survey respondents disclose more when computers ask the questions?" *Public opinion quarterly*, 77(4), 888-935.

Medway, R. L., and J. Fulton. "When more gets you less: A meta-analysis of the effect of concurrent web options on mail survey response rates." *Public Opinion Quarterly 76*.4 (2012): 733-746.

Nass, C., Y. Moon and P. Carney, 1999. "Are people polite to computers? Responses to computer-based interviewing systems." *Journal of Applied Social Psychology*, *29*(5), 1093-1109.

<u>Module 4.2: Interviews and Interviewers</u>

November 25: Recruitment and respondent selection; interviewing techniques (Suzer-Gurtekin)

Biemer, Paul P., and Lars E. Lyberg. 2003. "Chapter 5: Errors due to Interviewers and Interviewing." *Introduction to survey quality*. John Wiley & Sons: Hoboken, NJ, 149-187.

Optional readings:

Conrad, Frederick G., and Michael F. Schober. 2000. "Clarifying question meaning in a household telephone survey." *Public Opinion Quarterly,* 64(1), 1-28.

Conrad, Frederick G., et al. 2013. "Interviewer speech and the success of survey invitations." *Journal of the Royal Statistical Society: Series A,* 176(1), 191-210.

Gaziano, Cecilie. 2005. "Comparative analysis of within-household respondent selection techniques." *Public Opinion Quarterly,* 69(1), 124-157.

Schaeffer, Nora Cate, et al. 2013. "An interactional model of the call for survey participation: Actions and reactions in the survey recruitment call." *Public Opinion Quarterly,* 77(1), 323-351.

December 2: Interviewing techniques (con't); Interviewer effects (Suzer-Gurtekin)

Exercise 4 due
Final Exam available after class; Final Exam due by 4PM on December 9.

West, B. T., & Blom, A. G. (2017). Explaining interviewer effects: A research synthesis. *Journal of survey statistics and methodology*, *5*(2), 175-211.

Optional readings:

Davis, R. E. M.P. Couper, N.K. Janz, C.H. Caldwell, and K. Resnicow. 2010. "Interviewer Effects in Public Health Surveys." *Health Education Research, 25* (1), 14-26.

O'Muircheartaigh, Colm and Pamela Campanelli. 1998. "The relative impact of interviewer effects and sample design effects on survey precision." *Journal of the Royal Statistical Society: Series A,* 161(1), 63-77.

West, Brady. T., Frederick. G. Conrad, Frauke Kreuter and Felicitas Mittereder. 2018. "Can conversational interviewing improve survey response quality without increasing interviewer effects? "*Journal of the Royal Statistical Society: Series A,* 181(1), 181-203.

December 9: Final exam due by 4:00PM